

INDU 211: Introduction to Production and Manufacturing Systems

Term Paper Group Proposal

Department of Mechanical, Industrial and Aerospace Engineering · Concordia University

Term: Fall 2026 · Due Date: Thursday, October 8, 2026

1. Team Information

- **Team Name:** Team 4 — Production and Systems Group
 - **Course & Section:** INDU 211 / Section 2X
 - **Team Members & Responsibilities:**
 1. **Student Team Member** (Team Coordinator) — Project management, abstract, and introduction
 2. **Team Member 2** — Research on AI vs. Generative AI and workplace changes
 3. **Team Member 3** — Analysis of which industrial engineering areas can be automated
 4. **Team Member 4** — Common errors, hallucinations, and limitations of AI tools
 5. **Team Member 5** — The role of human engineers and essential skills for future graduates
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2. Paper Title

The Future of Industrial Engineering in the Generative AI Era: Opportunities, Limitations, and the Role of Human Engineers

3. Topic Overview & Objectives

Our team will investigate how Artificial Intelligence (AI) and Generative AI (like ChatGPT, Claude, and machine learning tools) are changing modern manufacturing and industrial engineering.

Industrial engineering focuses on making systems more efficient, reducing waste, and improving safety and quality. AI tools can now analyze huge amounts of factory data, suggest production schedules, and write code. However, AI also makes mistakes, hallucinates incorrect facts, and cannot understand real-world physical constraints on a factory floor.

Our paper directly addresses the five core questions assigned by the instructor:

1. **What are AI and Generative AI?** How do traditional AI models (which predict numbers or classify images) differ from Generative AI (which creates new text, designs, and code), and how are they used in workplaces today?
 2. **Which areas of Industrial Engineering can be automated?** We will compare tasks that AI can easily do (like generating initial production schedules or basic inventory forecasts) with tasks that still need humans (like factory floor layout adjustments, worker safety and ergonomics, and managing people).
 3. **What are the common errors and limitations of AI?** We will explore issues like "hallucinations" (AI making up incorrect answers), lack of physical common sense, data privacy risks when sharing factory data, and why AI cannot be held legally responsible for engineering decisions.
 4. **What is the role of the human engineer?** Why human judgment, common sense, and ethical responsibility are essential to supervise AI tools and handle unexpected problems on the production line.
 5. **What skills should engineering students learn?** What technical skills (like basic Python and data analysis) and soft skills (like problem-solving and communication) will help new graduates stand out in an AI-assisted job market.
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4. Proposed Report Outline (10-Page Term Paper)

- **1. Abstract / Executive Summary:** Short summary of our main findings.
- **2. Introduction:** What is industrial engineering, and what is the new wave of Generative AI?
- **3. How AI and GenAI Are Changing the Workplace:** Practical examples from manufacturing and logistics.
- **4. Automation in Industrial Engineering:** A clear breakdown of what can and cannot be automated.
- **5. The Big Limitations and Risks of AI:** Hallucinations, safety risks, and data privacy.
- **6. Why Human Engineers Are Irreplaceable:** Real-world judgment, ethics, and teamwork.

- **7. Future Skills for Engineering Graduates:** How students can prepare for the future.
 - **8. Conclusion:** Final thoughts on using AI as a tool rather than a replacement.
 - **9. References:** Academic papers from Google Scholar and ScienceDirect.
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5. Preliminary References

1. Bányai, T., et al. (2024). Generative artificial intelligence in smart manufacturing and logistics: A review. *Journal of Manufacturing Systems*, 74, 312–329.
 2. Lee, J., & Azamfar, M. (2024). Industrial AI and human-machine collaboration in engineering. *Computers & Industrial Engineering*, 188, 109880.
 3. Tao, F., & Zhang, M. (2023). Digital twins and smart manufacturing in modern industry. *IEEE Transactions on Industrial Informatics*, 19(11), 10842–10853.
 4. Institute of Industrial and Systems Engineers (IISE). (2025). The future of industrial engineering in the era of artificial intelligence. *IISE Solutions*, 57(2), 24–31.
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